

# NAYOUNG HA

nayoung@ucla.edu (607-319-9340)

## EDUCATION

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### University of California, Los Angeles

PhD, Computer Science, Advisor: Yuchen Cui

September 2026 - present

### Cornell University

MEng, Computer Science

August 2024 - May 2025

### Cornell University

BA, Computer Science

(1 year early graduation) August 2021- May 2024

## PUBLICATIONS

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Ruolin Ye, **Nayoung Ha**, Shuaixing Chen, Qiandao Liu, Gavin Chen, Shaoyang Stassen, Mark Zolotas, Jose Barreiros, Tapomayukh Bhattacharjee “RAG-Diff: Adapting Diffusion Policies to Dynamic Constraints with Retrieval-Augmented Guidance” *RSS 2026* [[arXiv](#)]

**Nayoung Ha\***, Ruolin Ye\*, Ziang Liu, Shubhangi Sinha, Tapomayukh Bhattacharjee “REPEAT: A Real2Sim2Real Approach for Pre-acquisition of Soft Food Items in Robot-assisted Feeding.” *IROS 2024* [[arXiv](#)]

## RESEARCH AND PROFESSIONAL EXPERIENCE

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### PoRTaL Lab

*Research Assistant*

June 2024 - May 2025

*Supervisor: Prof. Sanjiban Choudhury*

- Developed efficient ways of multi-view representation learning on Diffusion Policy
- Working on closing the domain gap between robot data and human data through guiding the policy with learning the object flow and determining robot feasible actions from mixed feasible and infeasible human video data.

### EmPRISE Lab

*Research Assistant*

February 2023 - Dec 2025

*Supervisor: Prof. Tapomayukh Bhattacharjee*

- Used Material Points Method simulation and assistive robots to resolve feeding task of soft body objects.
- Tested the Unity development of RCareWorld, a human-centric simulation world for physical and social robotic caregiving
- Implemented soft body simulation of RCareWorld on Unity

### Cornell University Autonomous Underwater Vehicle

*Software Engineer*

October 2021 - May 2024

*Supervisor: Prof. Hunter Adams*

- Finalist in Robonation’s 2024 International Robosub competition, placing 3rd out of 40 international teams
- Programmed vision modules using lab color space, filtering process, and image matching algorithms such as Scale-Invariant Feature Transform, Speed Up Robust Feature, and Oriented FAST and Rotated BRIEF technique.
- Created a Unity simulation for underwater pool environment enabling various mission tasks

### Big Care AI Lab

*Data Analyst Intern*

April 2021 - January 2022

*Supervisor: Prof. Min Song*

- Developed a food recommendation system considering ingredient preference and individual health factors for health care app company advised under Professor Min Song from Yonsei University

- Presented patent for food recommendation system regarding user preference using graph algorithms and matrix factorization
- Engaged in data analysis of nutrition factors and food consumption pattern of individuals using Pandas DataFrame and Scipy

## TEACHING EXPERIENCE AND MENTORING EXPERIENCE

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### **CS 4820 Introduction to Algorithms Head Teaching Assistant / Teaching Assistant**

*Cornell University*

*August 2023 - May 2025*

- Held office hours, answered Ed discussion questions, and graded assignments and exams.
- Helped students for topics including Gale-Shapley Algorithm, Greedy Algorithm, Dynamic Programming, Divide and Conquer, Network Flows, NP-completeness, Approximation Algorithms, Randomized Algorithms, Turing Machine, and Computability.

### **SoNIC (Software Defined Network Interface) Program Teaching Assistant**

*Cornell University*

*July 2023*

- Teaching assistant for SoNIC Workshop, a crash course in robotics with nearly 30 other U.S. students, hosted by the Cornell Ann S. Bowers College of Computing and Information Science and funded by NSF.
- Assisted in teaching students ROS, RViz, detection by ArUco markers and path searching algorithm such as A star.

### **CS 2110 Object-Oriented Programming and Data Structures Consultant**

*Cornell University*

*January 2023 - May 2023*

- Held office hours, assisted discussion sections, answered Ed discussion questions, and graded assignments and exams.
- Taught topics including Java syntax, Abstraction, Encapsulation, Testing, Interfaces, Inheritance, Dynamic dispatch, Exceptions, Bags, Generics Reading, Linked Structures, Ordered Collections, Efficiency, Recursion, Trees, Sorting, Hashing, GUIs, Concurrency, Synchronization, BFS, DFS, Dijkstra's Algorithm, Priority queues and heaps, and Iterators

### **Code Red Robotics Outreach Event**

*Ithaca High School*

*February 2023*

- Hosted an outreach event at Ithaca High School to raise interest in Robotics to our local community

## GRANTS AND AWARDS

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### **Bowers CIS Undergraduate Research Experience**

*Cornell University*

*June 2024 - August 2024*

- Received \$6000.00 for summer undergraduate research experience

### **Summer Experience Grant**

*Cornell University*

*May 2023 - August 2023*

- Received \$5370.00 for summer undergraduate research experience

### **Dean's List**

*Cornell University*

- Presented to Arts and Science students who have exemplary academic records